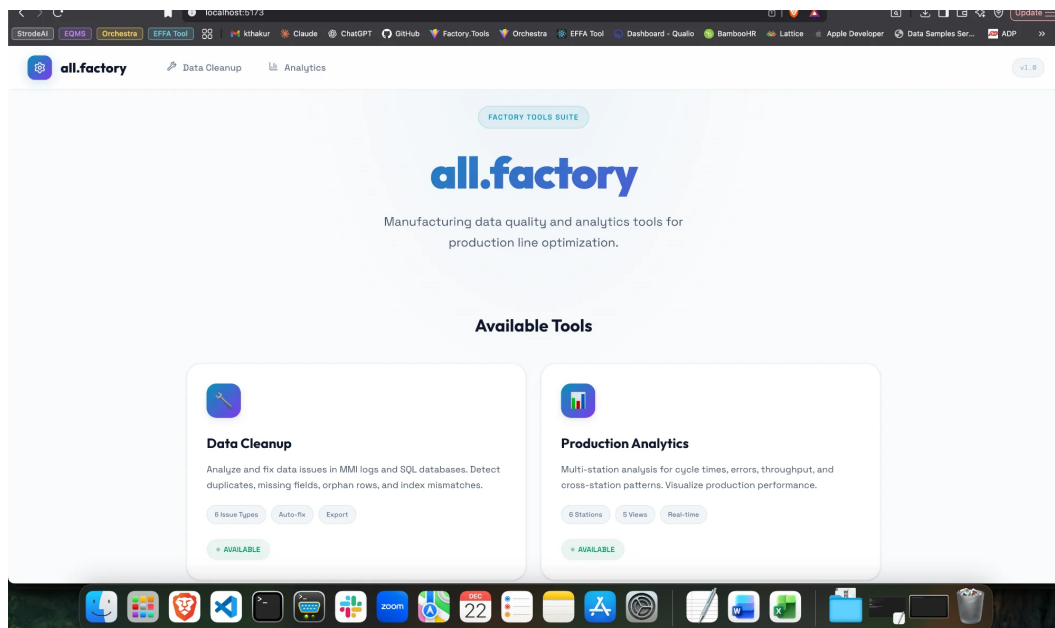


## FACTORY TOOLS SUITE

# all.factory

User Guide • Version 2.0

Manufacturing data quality and analytics tools  
for production line optimization.



■ Data Cleanup

■ Production Analytics

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# 1. Introduction

The all.factory suite is designed to help manufacturing teams identify and resolve data quality issues in production databases, while providing comprehensive analytics for production line optimization. It analyzes MMI (Man-Machine Interface) log files alongside SQL database exports to detect discrepancies, duplicates, missing data, and production patterns.

## Key Features

- **Data Cleanup:** Automated detection and correction of 6 issue types
- **Production Analytics:** Multi-station analysis with 5 specialized views
- **Cross-Station Analysis:** Error cascade and pattern detection
- **Serial Analysis:** Unit-by-unit cycle time tracking
- **Export Capabilities:** Download cleaned data and reports
- **Full Audit Trail:** MMI log evidence for every change

## Who Should Use This Tool

- Production engineers managing manufacturing data
- Quality assurance teams validating data integrity
- Database administrators performing routine cleanup
- Operations managers overseeing production line performance

## Supported Stations

| Station      | Code | Color  | File Types Accepted                |
|--------------|------|--------|------------------------------------|
| Bottom Shell | BS   | Indigo | Barcode Log, Error Log, SQL Export |
| Battery      | BA   | Green  | Barcode Log, Error Log, SQL Export |
| Trans        | TR   | Pink   | Barcode Log, Error Log, SQL Export |
| Top Shell    | TO   | Yellow | Barcode Log, Error Log, SQL Export |
| Laser        | LA   | Red    | Barcode Log, Error Log, SQL Export |
| FVT          | FV   | Cyan   | Barcode Log, Error Log, SQL Export |

## 2. Data Cleanup Tool

The Data Cleanup tool analyzes MMI log files and SQL database exports to automatically detect and help you fix common data quality issues.

### 2.1 Uploading Files

Upload your MMI log file and SQL export to begin analysis. Drag and drop files onto the upload zones or click to browse.

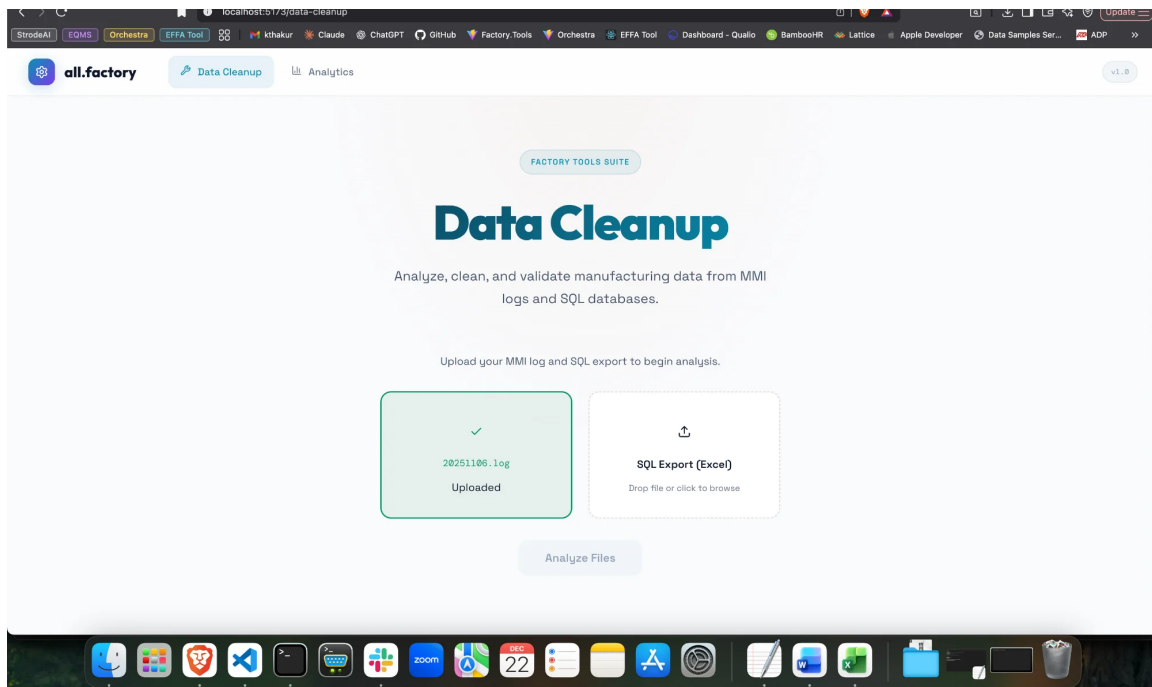


Figure 1: Data Cleanup upload screen

| File Type    | Description                                    | Formats     |
|--------------|------------------------------------------------|-------------|
| MMI Log File | Event log from your MMI system with timestamps | .log, .txt  |
| SQL Export   | Excel export of your production database table | .xlsx, .xls |



## 2.2 Understanding Analysis Results

After analysis completes, you'll see a statistics bar showing issue counts by type and status, plus a split-panel view with the issues list on the left and details on the right.

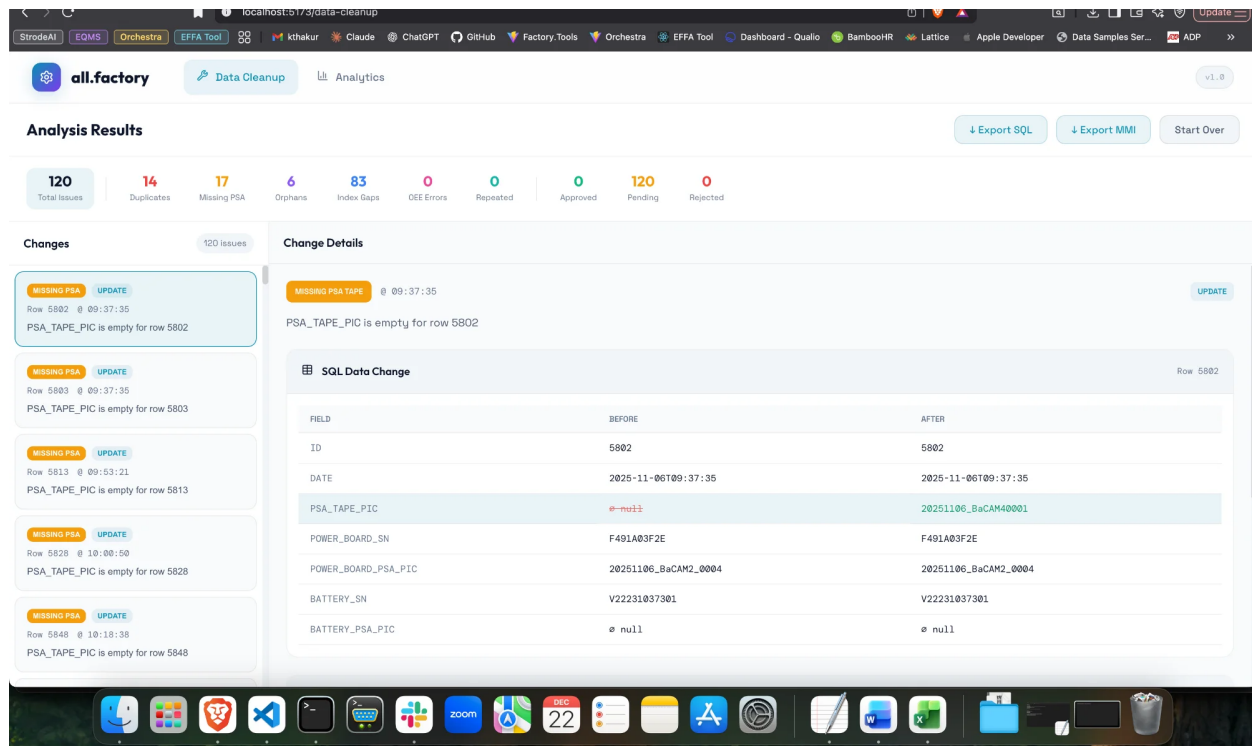


Figure 2: Analysis results with statistics bar and split-panel view

### Issue Types

| Issue Type       | Badge Color | Action | Description                                  |
|------------------|-------------|--------|----------------------------------------------|
| Duplicate Insert | Red         | DELETE | Same row inserted twice due to PLC trigger   |
| Missing PSA Tape | Orange      | UPDATE | PSA_TAPE_PIC field is empty but image exists |
| Orphan Row       | Purple      | FLAG   | Rows with PSA images but no serial numbers   |
| Index Mismatch   | Blue        | UPDATE | PSA image indices are incorrectly referenced |
| OEE Error        | Pink        | UPDATE | SQL/MMI error table discrepancies            |
| Repeated Insert  | Teal        | DELETE | Same content logged multiple times           |

## 2.3 Reviewing and Approving Changes

Click any issue in the left panel to see its details. The right panel shows the SQL data change (before/after comparison) and MMI log evidence supporting the proposed fix.

The screenshot displays the 'all.factory' Data Cleanup application interface. At the top, a navigation bar includes 'all.factory', 'Data Cleanup', and 'Analytics' tabs. Below this, the 'Analysis Results' section shows a summary of issues: 120 Total Issues, 14 Duplicates, 17 Missing PSA, 6 Orphans, 83 Index Gaps, 0 OEE Errors, 0 Repeated, 0 Approved, 120 Pending, and 0 Rejected. Buttons for 'Export SQL', 'Export MMI', and 'Start Over' are available. The 'Changes' section on the left lists 120 issues, with five items visible: Row 5802, Row 5803, Row 5813, Row 5828, and Row 5848, all marked as 'MISSING PSA' and 'UPDATE'. The 'Change Details' section on the right shows the details for Row 5802, including the 'BATTERY\_PSA\_PIC' field set to 'null'. Below this, the 'MMI Log Evidence' section displays three log entries: a timestamp '40 9:37:39 AM' with a value '+4,0,20251106\_BaCAM40001', a timestamp '38 9:37:35 AM' with a long SQL insert statement, and a timestamp '39 9:37:35 AM' with another long SQL insert statement. At the bottom of the details panel, there are 'Approve Change' and 'Reject' buttons. The interface is running on a local host, as indicated by the browser address bar.

Figure 3: Change details with SQL diff and MMI evidence

### Making Decisions

- Click **Approve Change** if the proposed modification is correct
- Click **Reject** if you disagree with the proposed change
- Approved changes show a green badge; rejected show red
- Use the statistics bar filters to find pending issues quickly

## 2.4 Exporting Cleaned Data

Once you've reviewed issues, use the export buttons in the header:

- **Export SQL:** Downloads cleaned\_data.xlsx with approved changes applied
- **Export MMI:** Downloads cleaned\_log.log with corrections noted
- Only **approved** changes are applied; rejected and pending are unchanged
- Click **Start Over** to reset and upload new files

## 3. Production Analytics

The Production Analytics tool provides multi-station analysis for cycle times, errors, throughput, and cross-station patterns across 5 specialized views.

### 3.1 Uploading Station Data

Upload log files for each station you want to analyze. Each station accepts three file types: Barcode Log, Error Log, and SQL Export. All files are optional.

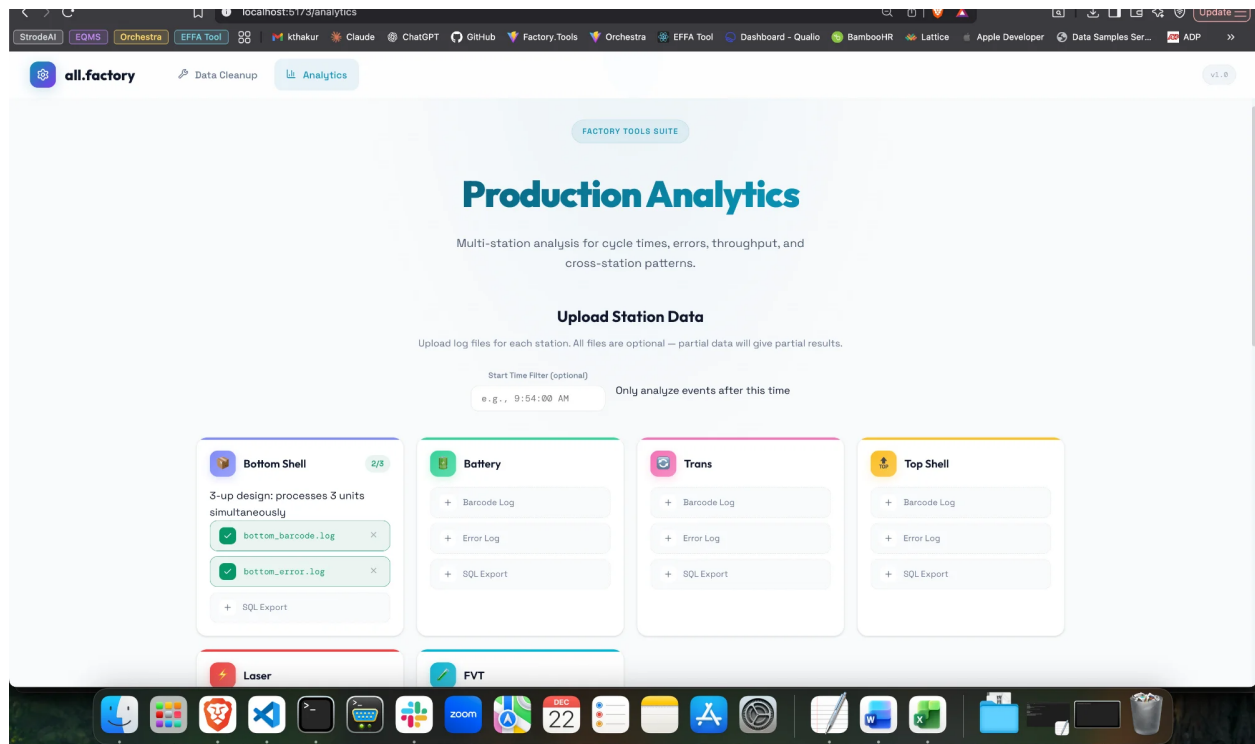


Figure 4: Station upload grid with files uploaded to Bottom Shell

Optionally set a **Start Time Filter** to analyze only events after a specific time. Click **Run Analysis** when ready.

## 3.2 Dashboard View

The Dashboard provides a high-level overview with summary cards showing total units, active stations, errors, downtime, and the bottleneck station. Below are individual station cards with key metrics.

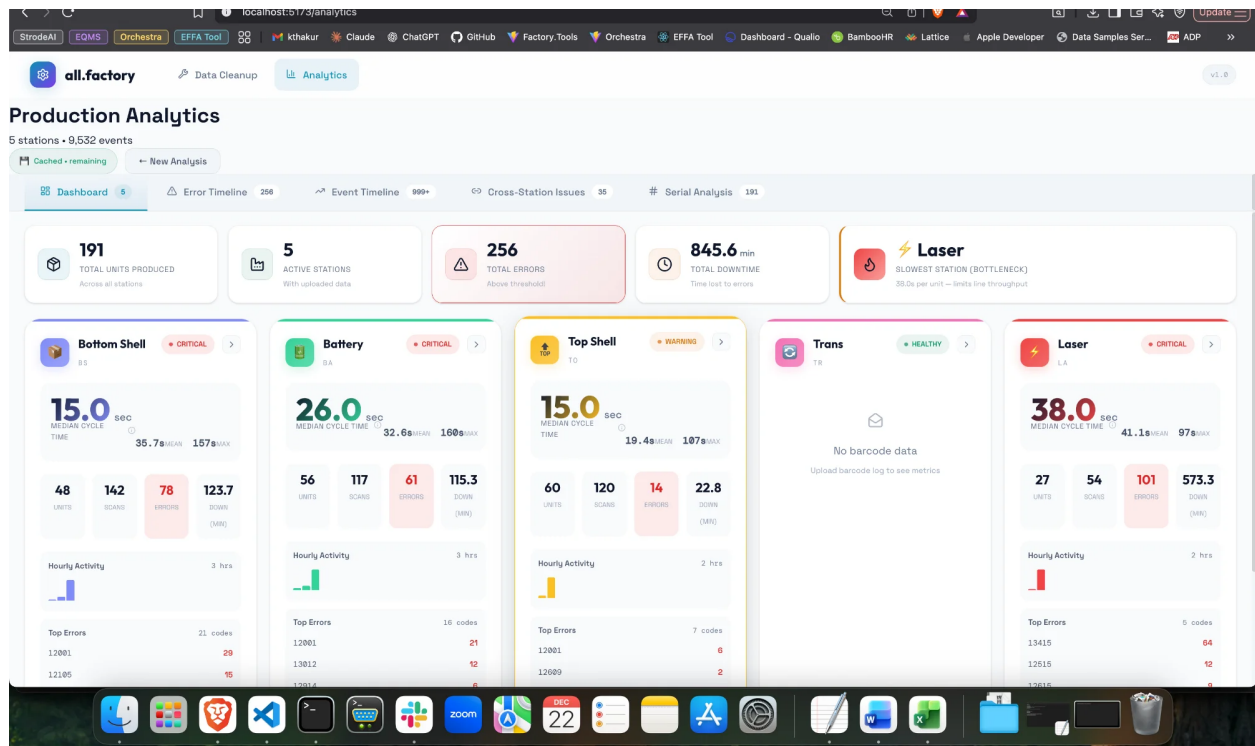


Figure 5: Dashboard with summary cards and station metrics

### Station Card Metrics

- **Health Status:** Healthy (green), Warning (yellow), or Critical (red)
- **Median Cycle Time:** Typical time between units
- **Units/Scans/Errors:** Key counts for the station
- **Downtime:** Time lost to errors in minutes
- **Hourly Activity:** Mini chart showing activity over time
- **Top Errors:** Most frequent error codes

### 3.3 Error Timeline

The Error Timeline shows when each error code occurs across all stations. Errors are grouped into swimlanes by station and code, with interactive filtering and zooming.

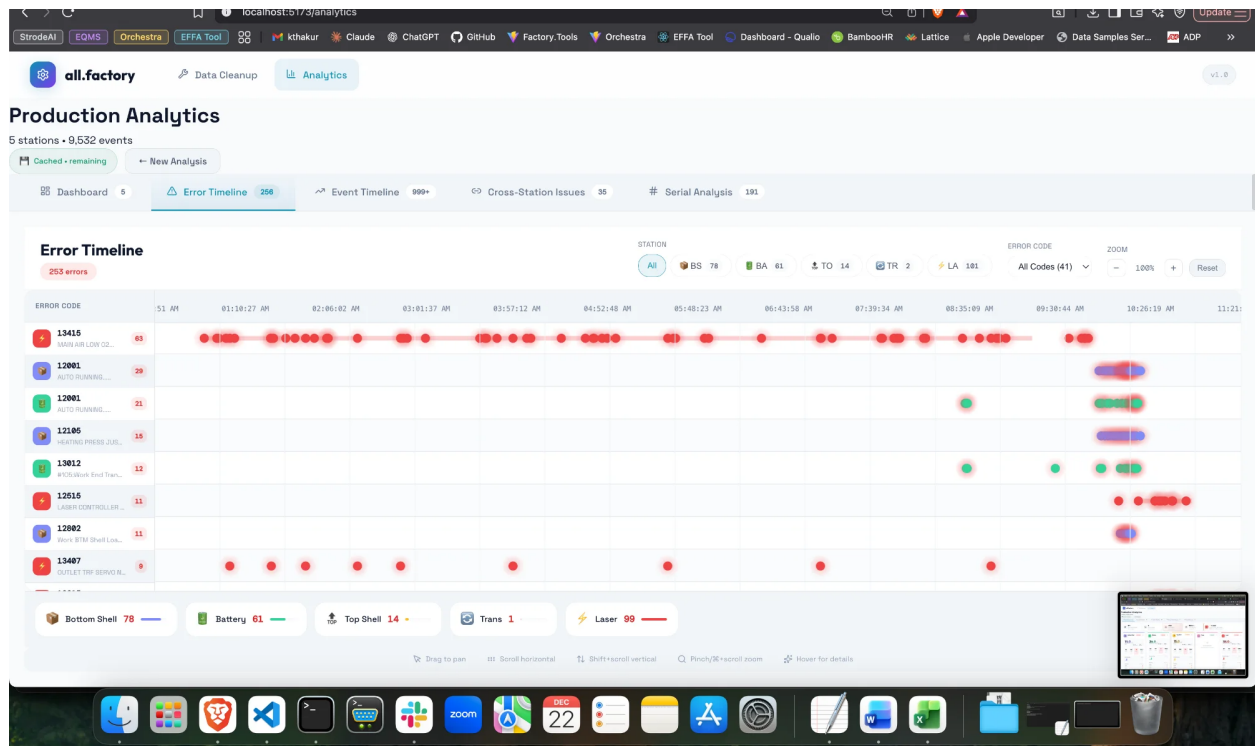


Figure 6: Error Timeline with swimlanes by error code

#### Controls

- **Station Filter:** Click station pills to show/hide specific stations
- **Error Code Filter:** Select a specific code from the dropdown
- **Zoom:** Use +/- buttons or pinch/■+scroll
- **Pan:** Drag to move, or scroll horizontally
- **Hover:** See error details including duration

## 3.4 Event Timeline

The Event Timeline shows all logged events (not just errors) with comprehensive filtering by station and category. Categories include Scan, Press, PSA, Database, System, PLC, Error, and Process.

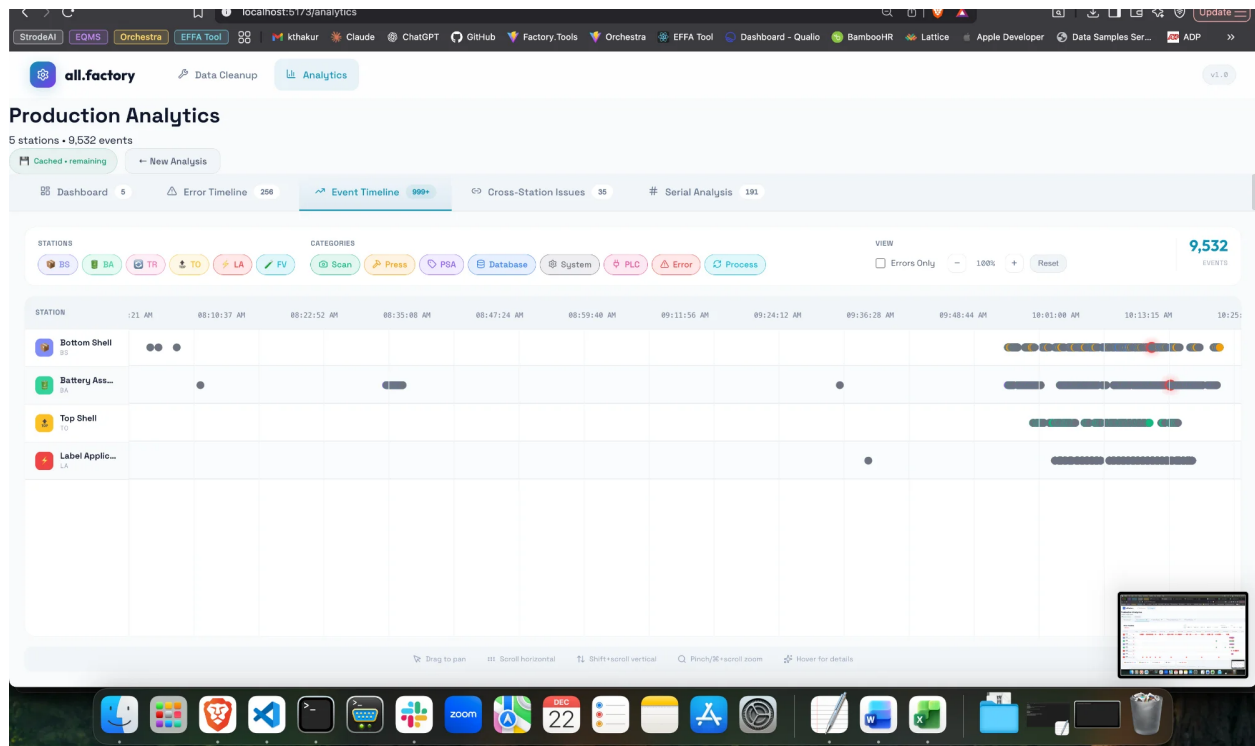


Figure 7: Event Timeline with category filters

Toggle category buttons to show/hide event types. Use "Errors Only" to focus on problem events. Hover over any event dot to see timestamp, station, and full event content.

## 3.5 Cross-Station Issues

This view analyzes patterns spanning multiple stations to identify systemic problems. It detects error cascades (errors propagating across stations) and recurring patterns.

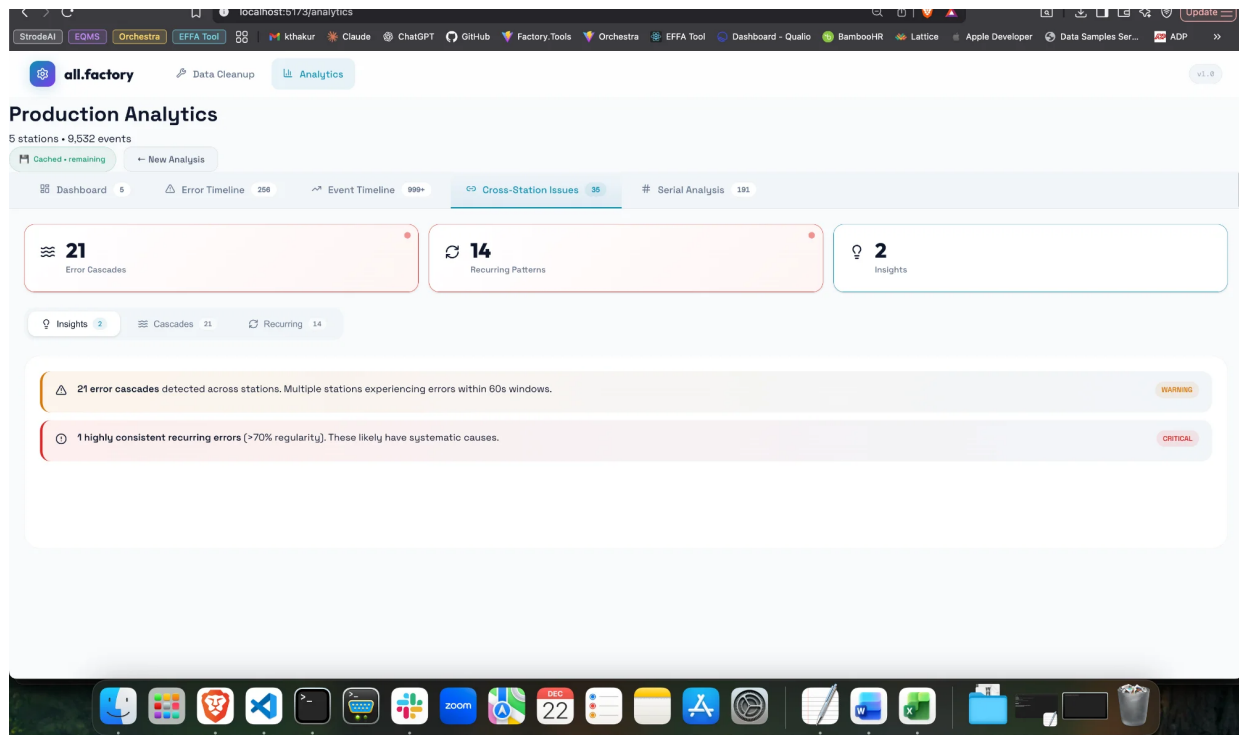


Figure 8: Cross-Station Issues with Insights tab

### Tabs

- **Insights:** AI-generated observations with severity levels
- **Cascades:** Errors that cascade across stations within 60s
- **Recurring:** Patterns with consistency scores



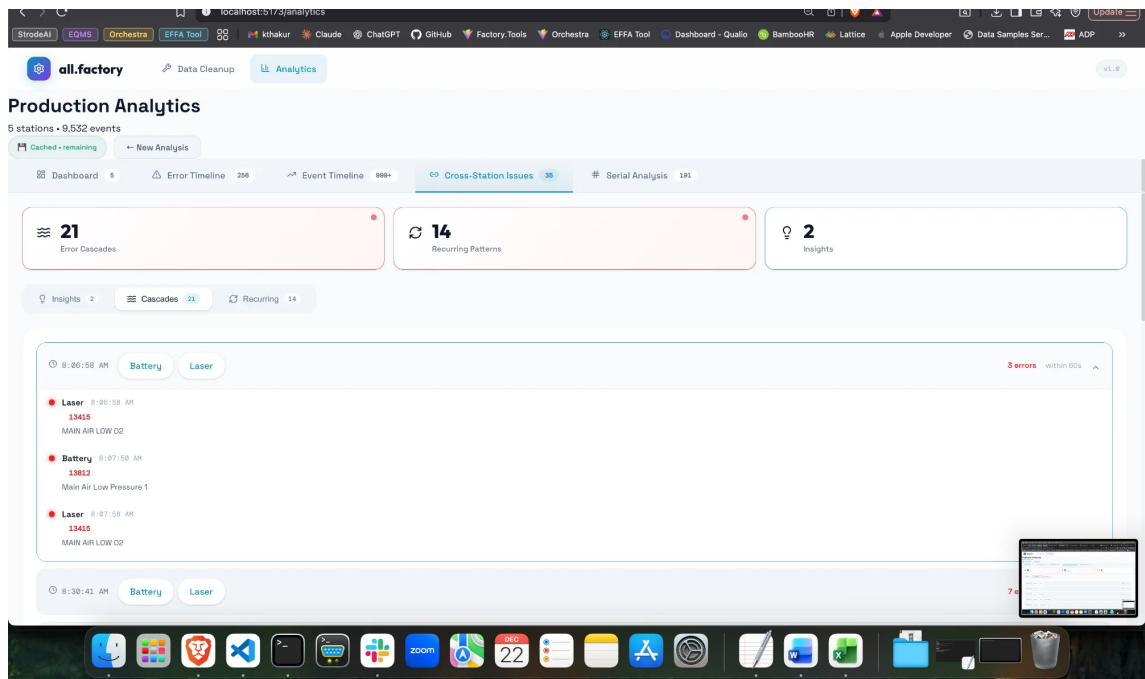


Figure 9: Expanded cascade showing error sequence across stations

## 3.6 Serial Analysis

The Serial Analysis view provides unit-by-unit cycle time analysis, showing gaps between consecutive units, identifying stoppages, and grouping production into runs.

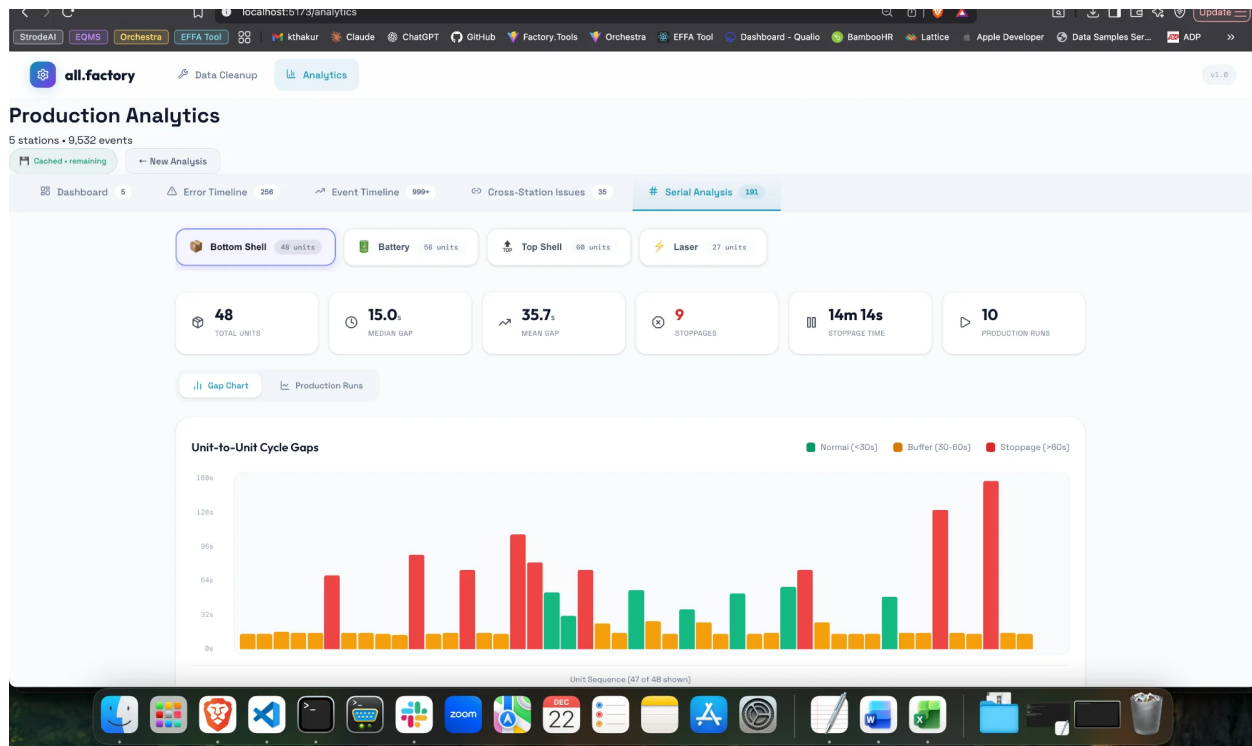


Figure 10: Serial Analysis with Gap Chart

### Gap Chart Legend

- **Green (Normal):** Gap < 30 seconds – healthy cycle time
- **Orange (Buffer):** Gap 30-60 seconds – slight delay
- **Red (Stoppage):** Gap > 60 seconds – production stoppage

Switch to **Production Runs** view for a Gantt-style timeline showing continuous production periods and stoppages, plus a table with run details including UPH (Units Per Hour).

## 3.7 Downloading Reports

Each analysis view can be exported as a standalone HTML report for sharing with team members who may not have access to the app. Click the **Export** button in the analytics header to open the download menu.

### Available Exports

| Export          | Contents                                                       | Interactivity                                              |
|-----------------|----------------------------------------------------------------|------------------------------------------------------------|
| All Reports     | Downloads all 5 reports at once                                | All features below                                         |
| Dashboard       | Summary cards, station health, KPIs, hourly charts, top errors | Static report                                              |
| Error Timeline  | Swimlane timeline of errors by station and code                | Station/code filters, zoom, pan, tooltips                  |
| Event Timeline  | Full event visualization across all stations                   | Station/category toggles, errors-only, zoom, pan, tooltips |
| Cross-Station   | Cascades, recurring patterns, insights                         | Tab switching, expandable cascades                         |
| Serial Analysis | Gap chart, Gantt timeline, runs table, units table             | Station tabs, gap/runs toggle, bar tooltips                |

### How to Export

1. Complete an analysis by uploading station data and clicking **Run Analysis**.
2. Click the **Export** button in the top-right header bar.
3. Select a specific view or choose **All Reports** to download all five at once.
4. Each report downloads as a self-contained **.html** file that opens in any browser.
5. Share via email, Slack, or shared drive. No server access needed to view.

### HTML Exports vs. In-App Views

The exported HTML reports reproduce the core layout, data, and key interactive features using plain HTML/CSS/JavaScript. Because they don't use the full React rendering pipeline, there are some differences:

| Aspect          | Details                                                                                                                                                                    |
|-----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visual fidelity | Closely matches in-app design but may have minor styling differences (spacing, shadows) since exports use static CSS rather than the component library.                    |
| Chart rendering | Timelines use inline SVG rather than dynamic canvas rendering. SVG charts are resolution-independent and print well, but hover interactions may feel slightly less smooth. |
| Data limits     | Gap charts cap at 200 bars; units table shows 50 rows max. In-app views handle larger datasets with virtualized scrolling.                                                 |
| Filtering       | All filter controls (station pills, category toggles, code dropdowns, errors-only checkbox) are fully functional. Count badges update dynamically as you filter.           |

*Tip: For the best interactive experience, use the in-app views. Use HTML exports to share results with stakeholders without app access, archive analysis snapshots, or view results offline.*

# 4. Reference Tables

## Analytics Tabs

| Tab                  | Purpose                                                           |
|----------------------|-------------------------------------------------------------------|
| Dashboard            | Summary KPIs, station health, cycle times, and throughput metrics |
| Error Timeline       | Interactive timeline showing when each error occurs by code       |
| Event Timeline       | Full event visualization with filtering by station and category   |
| Cross-Station Issues | Error cascades, recurring patterns, and correlations              |
| Serial Analysis      | Unit-by-unit cycle analysis with production runs                  |

## Issue Types

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| Laser        | LA   | Red    | Barcode Log, Error Log, SQL Export |
| FVT          | FV   | Cyan   | Barcode Log, Error Log, SQL Export |

# 5. Troubleshooting

## Upload fails or shows error:

- Verify the file format is correct (.log/.txt for MMI, .xlsx/.xls for SQL)
- Check that the file is not corrupted or empty
- Ensure the file size is under 50MB
- Try refreshing the page and uploading again

## Analysis takes too long:

- Files with over 10,000 rows may take 2-5 minutes
- Do not close or refresh the browser during analysis
- If analysis exceeds 10 minutes, contact your administrator

## No issues found:

- Your data may already be clean – this is good!
- Verify you uploaded the correct files for the time period
- Check that the MMI log and SQL export are from matching dates

## Export file won't open:

- Ensure you have Microsoft Excel or compatible application installed
- Try opening with Google Sheets or LibreOffice as an alternative
- Check your downloads folder for the file

## Analytics data is cached:

- Results are cached for 5 minutes to speed up tab switching
- Click 'New Analysis' to upload fresh data
- The cache indicator shows remaining time

## Getting Help

For assistance with server connectivity issues, permission or access problems, data format questions, or feature requests, contact your administrator.